MODULAR KOSZUL DUALITY: SECTIONS 2–4

RAEEZ LORGAT

Abstract. Sections 2–4 develop the curvature calculus of the modular bar construction. Section 2 installs the genus-g propagator via the prime form, computes the period defect, and identifies the single scalar $\kappa(\mathcal{A})$ controlling fibrewise curvature $d^2_{\mathrm{fib}} = \kappa \cdot \omega_g$. Section 3 develops the modular homotopy theory: the convolution L_∞-algebra $(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, \mu_r)$, the modular operad with Getzler–Kapranov Feynman transform, the six-component total differential, and the logarithmic Fulton–MacPherson compactification $\overline{\overline{\mathrm{FM}}}^{\log}$ of Mok 25 as ambient. Section 4 constructs the universal Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0 \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$, proves $[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ from $D^2_{\mathcal{A}} = 0$, and records the tridegree, depth, and convergence discipline. The five theorems are projections of this single universal object.

1. Curvature and the genus tower

At genus 0 the bar complex is flat: $d^2 = 0$ is an algebraic identity, equivalent to the Borcherds axiom. On a family of curves of genus $g \geq 1$, this nilpotence fails. The failure is controlled by a single scalar.

1.1. The prime form and the period defect.

On a compact Riemann surface Σ_g of genus $g \geq 1$, the function $z_1 - z_2$ has no global meaning: a meromorphic function with a single simple zero must have a simple pole. The *prime form* $E(z_1, z_2)$, a section of $K^{-1/2} \boxtimes K^{-1/2}$ vanishing to first order along the diagonal, replaces it. The genus-g propagator

$$\eta^{(g)}_{12} \; = \; d \log E(z_1, z_2) + \pi \sum_{\alpha,\beta=1}^{g} (\operatorname{Im} \Omega)^{-1}_{\alpha\beta}\, \omega_\alpha(z_1)\, \bar{\omega}_\beta(z_2) \tag{1.1}$$

is single-valued; the period correction couples to the Hodge bundle $\mathbb{E} = R^0 \pi_* \omega_\pi$ through the period matrix Ω.

Three properties of (1.1) constrain the genus tower. First, the logarithmic derivative $d \log E$ has conformal weight 1 in both variables, regardless of the conformal weight of the field being sewed: every edge of every graph sum at every genus uses the standard Hodge bundle $\mathbb{E}_1$. Second, the period correction is harmonic in each variable and involves $(\operatorname{Im} \Omega)^{-1}$, the Arakelov Green function on Σ_g. Third, the Arnold relation acquires a defect proportional to the Arakelov form, and this defect propagates into the bar differential.

1.2. Curvature.

Let $\pi : \mathcal{C}_g \to \overline{\mathcal{M}}_g$ be the universal curve. On each fibre Σ_g, the propagator $\eta^{(g)}$ is single-valued but no longer closed: the period correction has a $\bar{\partial}$-component, and the Arnold relation fails. The fibrewise bar differential satisfies

$$d^2_{\mathrm{fib}} \; = \; \kappa(\mathcal{A}) \cdot \omega_g \,, \tag{1.2}$$

where ω_g is the Arakelov $(1,1)$-form on the fibre and $\kappa(\mathcal{A}) \in \mathbf{C}$ is the *modular characteristic*. The modular characteristic is intrinsic to the chiral algebra; it is determined by the leading OPE singularity and is visible already at genus 0 as the Σ_2-coinvariant scalar shadow of the collision residue: for abelian

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and Virasoro-type families $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$, while for non-abelian affine Kac–Moody in the trace-form convention $\mathrm{av}(r(z)) = \kappa_{\mathrm{dp}}(\mathcal{A})$ and $\kappa(\mathcal{A}) = \kappa_{\mathrm{dp}}(\mathcal{A}) + \dim(\mathfrak{g})/2$.

The failure of $d^2 = 0$ at genus $g \geq 1$ is not a defect of the construction; it is the construction detecting global geometry. Period integrals restore nilpotence: the Gauss–Manin connection absorbs the fibrewise curvature, and the total differential satisfies $D_g^2 = 0$. The curvature (1.2) is the infinitesimal avatar of this restoration.

1.3. **The genus expansion.** For a modular Koszul chiral algebra $\mathcal{A}$ with a single generator (or more generally with all generators of the same conformal weight), the genus-g obstruction class $\mathrm{obs}_g(\mathcal{A})$ is proportional to the top Chern class of the Hodge bundle:

$$\mathrm{obs}_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g\,, \qquad g \geq 1\,.$$

The free energy $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$, where

$$\lambda_g^{\mathrm{FP}} \;=\; \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$$

is the Faber–Pandharipande coefficient (B_{2g} the $2g$-th Bernoulli number), packages the entire genus tower into the Hirzebruch $\hat{A}$-class:

$$\sum_{g=1}^{\infty} F_g(\mathcal{A}) \quad {}^{2g-2} \;=\; \frac{\kappa(\mathcal{A})}{2}\big[\hat{A}(i \quad) - 1\big]\,, \qquad \hat{A}(x) = \frac{x/2}{\sinh(x/2)}\,. \tag{1.3}$$

At genus 1: $\lambda_1^{\mathrm{FP}} = 1/24$. At genus 2: $\lambda_2^{\mathrm{FP}} = 7/5760$. The universal ratio $F_2/F_1 = 7/240$ is independent of $\mathcal{A}$. The series converges for $|\quad| < 2\pi$, the first zero of $\sin(\quad/2)$.

The genus-1 identity $\mathrm{obs}_1 = \kappa \cdot \lambda_1$ is unconditional: it holds for all modular Koszul chiral algebras, regardless of the number or weights of their generators. At genus $g \geq 2$, the story is more delicate.

Remark 1.1 (Multi-weight correction). For a chiral algebra with generators of distinct conformal weights $h_1, \ldots, h_r$, the genus-g free energy decomposes as

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \;+\; \delta F_g^{\mathrm{cross}}(\mathcal{A})\,,$$

where $\delta F_g^{\mathrm{cross}}$ is a graph sum over mixed-channel boundary graphs of $\overline{\mathcal{M}}_{g,0}$. At genus 1, $\delta F_1^{\mathrm{cross}} = 0$ unconditionally. For uniform-weight algebras ($h_1 = \cdots = h_r$), $\delta F_g^{\mathrm{cross}} = 0$ at all genera. For multi-weight algebras at $g \geq 2$, the correction is generically nonzero: for $\mathcal{W}_3$ at genus 2, $\delta F_2^{\mathrm{cross}}(\mathcal{W}_3) = (c+204)/(16c) > 0$ for all $c > 0$.

1.4. **The modular characteristic.** The modular characteristic $\kappa(\mathcal{A})$ is the single scalar controlling the genus tower. It is computed from the leading OPE singularity and fixed universally by the genus-1 curvature extracted through the bar construction.

The modular characteristic is additive under independent sums ($\kappa(\mathcal{A} \oplus \mathcal{A}') = \kappa(\mathcal{A}) + \kappa(\mathcal{A}')$ when the mixed OPE vanishes) and constrained by Koszul duality: $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = 0$ for Kac–Moody and free-field algebras; $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$ for Virasoro. In general, $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = \rho \cdot K$ for $\mathcal{W}$-algebras, where ρ is the anomaly ratio and K the complementarity constant.

At the self-dual point ($c = 13$ for Virasoro), the complementarity sum is symmetric: $\kappa(\mathrm{Vir}_{13}) = 13/2 = \kappa(\mathrm{Vir}_{13}^!)$. At the critical level ($k = -h^\vee$ for Kac–Moody), $\kappa = 0$ and the bar complex is flat at all genera: the content migrates from curvature to bar cohomology, where it appears as the Feigin–Frenkel centre.

2. The five main theorems

The bar construction of a chiral algebra on a curve produces a factorization coalgebra on the Ran space of the curve. The five main theorems describe what this coalgebra determines: the Koszul dual,

TABLE 1. Modular characteristic for the standard families

Algebra $\mathcal{A}$	$\kappa(\mathcal{A})$	**Shadow depth class**
Heisenberg $\mathcal{H}_k$	k	**G**
Affine Kac–Moody $\widehat{\mathfrak{g}}_k$	$\dfrac{\dim(\mathfrak{g})\cdot(k+h^\vee)}{2h^\vee}$	**L**
$\beta\gamma$ system	$c/2$	**C**
Virasoro Vir_c	$c/2$	**M**
$\mathcal{W}_3^k(\mathfrak{sl}_3)$	$5c/6$	**M**
$\mathcal{W}_N^k(\mathfrak{sl}_N)$	$c\cdot\sum_{j=2}^{N}\frac{1}{j}$	**M**

the original algebra, the complementarity decomposition, the genus tower, and the chiral Hochschild cohomology. Each theorem is a projection of the universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ (Section 3).

2.1. Theorem A: Verdier intertwining.

Theorem 2.1 (A). *Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X. There is an adjunction*

$$B\ :\ \mathrm{Alg}_{\mathrm{aug}}(\mathrm{Fact}(X))\ \rightleftarrows\ \mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X))\ :\ \Omega$$

between augmented factorization algebras and conilpotent factorization coalgebras, with unit and counit the universal twisting morphisms. Verdier duality on $\mathrm{Ran}(X)$ *intertwines the bar of $\mathcal{A}$ with the bar of its Koszul dual:*

$$\mathbb{D}_{\mathrm{Ran}}\big(B(\mathcal{A})\big)\ \simeq\ B(\mathcal{A}^!)\,, \tag{2.1}$$

as factorization coalgebras, or equivalently $\mathbb{D}_{\mathrm{Ran}}(B(\mathcal{A}))\simeq\mathcal{A}^!_\infty$ *as the homotopy Koszul dual algebra.*

The adjunction is formal: it requires only the augmented factorization-algebra structure and the duality on $\mathrm{Ran}(X)$. The intertwining (2.1) is a deeper statement: it says that the Verdier dual of the bar *coalgebra* is the bar of the Koszul dual *algebra*. The identification of the homotopy Koszul dual $\mathcal{A}^!_\infty$ (retaining full chain-level data) with the strict Koszul dual $\mathcal{A}^! = (H^*(B(\mathcal{A})))^\vee$ (retaining only cohomology) is the content of Theorem A on the Koszul locus, not a tautology.

2.2. Theorem B: bar-cobar inversion.

Theorem 2.2 (B). *On the Koszul locus, the cobar of the bar recovers the original algebra:*

$$\Omega\big(B(\mathcal{A})\big)\ \xrightarrow{\sim}\ \mathcal{A}\,.$$

The counit of the bar-cobar adjunction is a quasi-isomorphism of augmented factorization algebras. At genus 0 *this is unconditional. For arbitrary modular pre-Koszul inputs at $g\geq 1$, the extension uses the modular axiom; on the standard CFT-type landscape it is unconditional except for integer-spin $\beta\gamma$.*

Bar-cobar inversion is a round-trip, not a duality. It does not produce the Koszul dual (that is Verdier duality, Theorem A); it does not produce the bulk algebra (that is the chiral derived centre, Theorem H). It recovers $\mathcal{A}$ itself.

The proof proceeds by constructing a contracting homotopy for the counit, using the PBW filtration as a control parameter: the associated graded of the cobar complex coincides with a cofree resolution of the PBW-graded algebra, and the filtration spectral sequence degenerates at E_2. Convergence of the spectral sequence requires the Koszul hypothesis; for non-Koszul algebras, bar-cobar inversion requires completion.

2.3. Theorem C: complementarity.

Theorem 2.3 (C)**.** *For a Koszul pair $(\mathcal{A}, \mathcal{A}^!)$ on X and each $g \geq 0$, the complementarity complex*

$$\mathbf{C}_g(\mathcal{A}) \;:=\; R\Gamma\big(\overline{\mathcal{M}}_g,\, \mathcal{Z}_{\mathcal{A}}\big)$$

decomposes into homotopy eigenspaces of the Verdier involution σ:

$$\mathbf{C}_g \;\simeq\; \mathbf{Q}_g(\mathcal{A}) \;\oplus\; \mathbf{Q}_g(\mathcal{A}^!)\,.$$

For $g \geq 1$, the summands $\mathbf{Q}_g(\mathcal{A}) = \mathrm{fib}(\sigma - \mathrm{id})$ and $\mathbf{Q}_g(\mathcal{A}^!) = \mathrm{fib}(\sigma + \mathrm{id})$ are Lagrangian for the (-1)-shifted symplectic pairing on $\mathbf{C}_g$.

The eigenspace decomposition (C1) is unconditional: it holds for all modular Koszul algebras at all genera. The scalar identity $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ on each summand (C2) requires the uniform-weight hypothesis at $g \geq 2$. The Lagrangian property, a (-1)-shifted symplectic statement in the sense of Pantev–Toën–Vaquié–Vezzosi, is conditional on perfectness and nondegeneracy of the cyclic pairing.

2.4. Theorem D: the modular characteristic.

Theorem 2.4 (D)**.** *For a uniform-weight modular Koszul chiral algebra $\mathcal{A}$, the genus-g obstruction class satisfies* $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ (UNIFORM-WEIGHT) *for all $g \geq 1$, where $\kappa(\mathcal{A})$ is a universal, additive, duality-constrained scalar fixed by the genus-1 curvature and already visible in the leading OPE singularity of $\mathcal{A}$. The generating function is the Hirzebruch $\hat{A}$-class* ((1.3))*. At genus 1, the identity is unconditional: it holds for all modular Koszul algebras regardless of weight structure. At $g \geq 2$ for multi-weight algebras, the full free energy acquires the explicit cross-channel correction $\delta F_g^{\mathrm{cross}}$.*

The universality is the point: no matter how complicated the OPE structure, the genus tower is controlled by a single number. Theorem D is the all-genera identity on the uniform-weight lane; at genus 1 it is unconditional (Remark 1.1).

2.5. Theorem H: chiral Hochschild cohomology.

Theorem 2.5 (H)**.** *For the generic Virasoro and principal $\mathcal{W}$-algebra families, the chiral Hochschild cohomology is concentrated in degrees $\{0, 1, 2\}$ with Hilbert polynomial $P(t) = 1 + t^2$. For affine Kac–Moody at generic level,*

$$\mathrm{ChirHoch}^1(V_k(\mathfrak{g})) \cong \mathfrak{g}.$$

At critical level, the cohomology is identified with the continuous Lie algebra cohomology by the Beilinson–Drinfeld comparison theorem.

The chiral derived centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ is the universal bulk algebra: the algebra of operators in the interior that act on the boundary algebra $\mathcal{A}$. Theorem H constrains its cohomology. For the Virasoro algebra, the total dimension of $\mathrm{ChirHoch}^*(\mathrm{Vir}_c)$ is at most 4; this is a finite-dimensionality statement, not a polynomial-ring statement.

2.6. **The Koszulness meta-theorem.** The hypothesis shared by Theorems A–H is chiral Koszulness: bar cohomology $H^*(B(\mathcal{A}))$ concentrated in bar degree 1. The meta-theorem characterizes this condition in twelve equivalent ways, ten of which are unconditional, one conditional, one partial.

Theorem 2.6 (Koszulness characterization)**.** *For a chiral algebra $\mathcal{A}$ on X with PBW filtration, conditions* (i)–(x) *are equivalent. Under additional perfectness hypotheses,* (xi) *is equivalent. Condition* (xii) *implies* (x)*; the converse is open.*

(i) *$\mathcal{A}$ is chirally Koszul.*
(ii) *The PBW spectral sequence collapses at E_2.*
(iii) *The minimal A_∞-model of $B(\mathcal{A})$ is formal: $m_n = 0$ for $n \geq 3$.*
(iv) *Ext diagonal vanishing:* $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X) = 0$ *for $p \neq q$.*
(v) *The bar-cobar counit $\Omega(B(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism.*
(vi) *The Barr–Beck–Lurie comparison is an equivalence.*
(vii) *Factorization homology $\int_{\Sigma_g} \mathcal{A}$ is concentrated in degree 0 for all g.*
(viii) $\mathrm{ChirHoch}^*(\mathcal{A})$ *is polynomial in degrees* $\{0, 1, 2\}$.
(ix) *The Kac–Shapovalov determinant is nonzero in the bar-relevant range.*
(x) *Restriction to every FM boundary stratum is acyclic outside degree* 0.
(xi) *Lagrangian criterion* (conditional).
(xii) *$\mathcal{D}$-module purity* (one-directional: (x) $\Rightarrow$ (xii)).

The circuit of unconditional equivalences runs through five independent mathematical domains: PBW degeneration (commutative algebra), A_∞-formality (homotopical algebra), Barr–Beck–Lurie (higher categories), Kac–Shapovalov (representation theory), and FM boundary acyclicity (algebraic topology). That these five conditions define the same class of algebras is the depth of the meta-theorem.

3. The universal Maurer–Cartan element and the shadow tower

The five theorems of Section 2 are projections of a single algebraic object: the universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ of the modular convolution algebra.

3.1. The bar-intrinsic construction. The genus-completed bar differential $D_{\mathcal{A}} = \sum_{g \geq 0} \hbar^g d^{(g)}_{\mathcal{A}}$ satisfies $D^2_{\mathcal{A}} = 0$ by the codimension-2 face cancellation on $\overline{\mathcal{M}}_{g,n}$ (the same mechanism as §??, extended to all genera by Mok's log Fulton–MacPherson compactification). Write $d_0 = d^{(0)}_{\mathcal{A}}$ for the genus-0 bar differential.

Construction 3.1 (Bar-intrinsic MC element)**.** Define the positive-genus correction

$$\Theta_{\mathcal{A}} \; := \; D_{\mathcal{A}} - d_0 \; = \; \sum_{g \geq 1} \hbar^g d^{(g)}_{\mathcal{A}} \; \in \; \prod_{g \geq 1} \mathrm{CoDer}^{\mathrm{cyc}}\big(B^{(g)}_X(\mathcal{A})\big)[1] \, .$$

Theorem 3.2. *$\Theta_{\mathcal{A}}$ is a Maurer–Cartan element of the modular cyclic deformation complex:*

$$d_0(\Theta_{\mathcal{A}}) + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] \; = \; 0 \, . \tag{3.1}$$

Proof sketch. Expand $D^2_{\mathcal{A}} = (d_0 + \Theta_{\mathcal{A}})^2 = 0$. Since $d_0^2 = 0$ (genus-0 bar flatness), the remaining terms give $d_0(\Theta_{\mathcal{A}}) + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$, which is the MC equation. The bracket is the convolution bracket on the modular cyclic deformation complex, inherited from stable-curve gluing; its square-zero property is a formal consequence of $d_0^2 = 0$. □

The construction requires no restriction to simple Lie symmetry, no one-channel hypothesis, and no tautological-line support condition. It works for all modular Koszul chiral algebras. The modular characteristic κ, the cubic shadow $\mathfrak{C}$, and the quartic resonance class $\mathfrak{Q}$ are degree projections of this single element:

$$\kappa(\mathcal{A}) = \pi_2(\Theta_{\mathcal{A}}) \, , \qquad \mathfrak{C}(\mathcal{A}) = \pi_3(\Theta_{\mathcal{A}}) \, , \qquad \mathfrak{Q}(\mathcal{A}) = \pi_4(\Theta_{\mathcal{A}}) \, .$$

Each degree projection carries more information than the last; the full $\Theta_{\mathcal{A}}$ carries all of it.

3.2. **The shadow obstruction tower.** The degree-r projection $\mathrm{Sh}_r(\mathcal{A}) := \pi_r(\Theta_{\mathcal{A}})$ is the r-th shadow coefficient. The sequence $S_2 = \kappa$, $S_3 = \alpha$, $S_4, S_5, \ldots$ constitutes the *shadow obstruction tower*. On any one-dimensional primary slice L of the cyclic deformation complex, the MC equation (3.1) imposes a quadratic constraint.

Theorem 3.3 (Riccati algebraicity). *The weighted shadow generating function $H(t) := \sum_{r\geq 2} r\, S_r\, t^r$ satisfies*

$$H(t)^2 \;=\; t^4\, Q_L(t)\,,$$

where

$$Q_L(t) \;=\; (2\kappa + 3\alpha t)^2 \;+\; 2\Delta\, t^2\,, \qquad \Delta := 8\kappa S_4\,,$$

is the shadow metric*: a quadratic polynomial in t determined by three genus-0 invariants (κ, α, S_4). The entire shadow obstruction tower on L is determined by Q_L.*

The generating function $H(t) = t^2\sqrt{Q_L(t)}$ is algebraic of degree 2. The shadow obstruction tower is neither a Taylor expansion that might diverge nor a formal power series requiring analytic continuation: it is the expansion of an explicit algebraic function. The MC equation, which a priori is an infinite system of coupled nonlinear equations, reduces on each primary slice to a single quadratic identity.

3.3. **The G/L/C/M classification.** The discriminant $\Delta = 8\kappa S_4$ of the shadow metric forces a universal dichotomy.

Definition 3.4 (Shadow depth classification). The *shadow depth* of $\mathcal{A}$ is $r_{\max}(\mathcal{A}) := \sup\{r \geq 2 : S_r \neq 0\}$. The shadow depth class is:

Class	Name	$r_{\max}$	Mechanism
G	Gaussian	2	Q_L perfect square, $\alpha = 0$
L	Lie	3	Q_L perfect square, $\alpha \neq 0$
C	Contact	4	stratum separation
M	Mixed	∞	$\Delta \neq 0$: Q_L irreducible

The mechanism: when $\Delta = 0$, the shadow metric is a perfect square $Q_L(t) = (2\kappa + 3\alpha t)^2$ and $\sqrt{Q_L}$ is polynomial: the tower terminates. If $\alpha = 0$ it terminates at degree 2 (class **G**: Heisenberg); if $\alpha \neq 0$ at degree 3 (class **L**: affine Kac–Moody). When $\Delta \neq 0$, $\sqrt{Q_L}$ is irrational: the tower runs forever (class **M**: Virasoro, $\mathcal{W}_N$). Class **C** ($\beta\gamma$) escapes the dichotomy by stratum separation: the quartic contact invariant lives on a charged stratum whose self-bracket exits the complex by rank-one rigidity.

The four classes are a homotopy-invariant classification: they coincide with the L_∞-formality depth of the transferred minimal model. Class **G** is L_∞-formal (all higher brackets vanish). Class **L** has exactly one nontrivial Massey product. Class **C** has formality obstructions through degree 4. Class **M** is intrinsically non-formal: no finite truncation of the L_∞ structure is quasi-isomorphic to a dg Lie algebra.

Archetypes:

Class	Archetype	κ	Δ
G	Heisenberg $\mathcal{H}_k$	k	0
L	$\widehat{\mathfrak{g}}_k$	$\dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$	0
C	$\beta\gamma$	$c/2$	0 (stratum separation)
M	Vir_c	$c/2$	$40/(5c+22)$

For the Virasoro algebra, the quartic contact invariant is $Q^{\mathrm{contact}}_{\mathrm{Vir}} = 10/[c(5c+22)]$ and the critical discriminant is $\Delta_{\mathrm{Vir}} = 8 \cdot (c/2) \cdot 10/[c(5c+22)] = 40/(5c+22)$. Every shadow coefficient S_r for $r \geq 5$ is a rational function of c with poles only at $c = 0$ and $c = -22/5$:

$$S_5 = -\frac{48}{c^2(5c+22)}\,, \qquad S_6 = \frac{80(45c+193)}{3\,c^3(5c+22)^2}\,, \qquad S_7 = -\frac{2880(15c+61)}{7\,c^4(5c+22)^2}\,.$$

These are the fingerprints of a class-**M** algebra: the shadow tower is infinite, alternating in sign, and controlled by an irrational algebraic function.

3.4. **The gauge-gravity dichotomy.** The classification has a physical interpretation. At $\kappa = 0$ the bar complex is flat at all genera: the genus tower carries no curvature, the free energy vanishes, and the content lives in bar cohomology. This is the regime of gauge theory: the Feigin–Frenkel centre (the Langlands dual of the affine algebra at critical level) is the genus-0 bar cohomology $H^0(B(\widehat{\mathfrak{g}}_{-h^\vee}))$.

At $\kappa \neq 0$ the bar complex is curved: every genus contributes a nonzero obstruction class, the genus tower is nontrivial, and the free energy is the $\hat{A}$-genus. This is the regime of gravity: the bar curvature is the perturbative genus expansion.

The Koszul duality $\mathcal{A} \leftrightarrow \mathcal{A}^!$ exchanges gauge and gravity sectors. For affine Kac–Moody at generic level, $\kappa(\widehat{\mathfrak{g}}_k) \neq 0$ and the algebra is curved; at the Koszul-dual critical level $k' = -k - 2h^\vee$, $\kappa(\widehat{\mathfrak{g}}_{k'}) \neq 0$ as well, and both sides are gravitational. The gauge regime (critical level, $\kappa = 0$) is the fixed locus of a one-parameter family.

For the Virasoro algebra, the Koszul dual is $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$. The gauge locus is $c = 0$ (topological: $\kappa = 0$); the gravitational locus is $c \neq 0$. The self-dual point $c = 13$ ($\kappa = 13/2 = \kappa^!$) is the critical string. The Clifford algebra of the complementarity pairing degenerates at $c = 26$ ($\kappa(\mathrm{Vir}_0^!) = 0$): the 26-dimensional bosonic string is the critical point where the Koszul dual becomes flat.

The dichotomy gauge/κ=0 vs gravity/$\kappa \neq 0$ is not a physical input: it is a theorem about the bar complex. The physical interpretation follows from the identification of bar curvature with perturbative genus expansions in quantum field theory.

Perimeter Institute for Theoretical Physics, Waterloo, ON N2L 2Y5, Canada
Email address: rlorgat@perimeterinstitute.ca